

Q1) [10 marks]

- a) Determine the state space matrices A,B,C & D for a system given

$$Y'''(t) + 0.6Y''(t) + 2Y = 3r(t)$$

Parameters

A:

B:

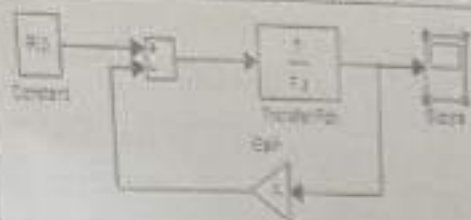
C:

D:

- b) Find the transfer function of above equation

Q1. Plot the output response for each of the following conditions

(7.5 mark)



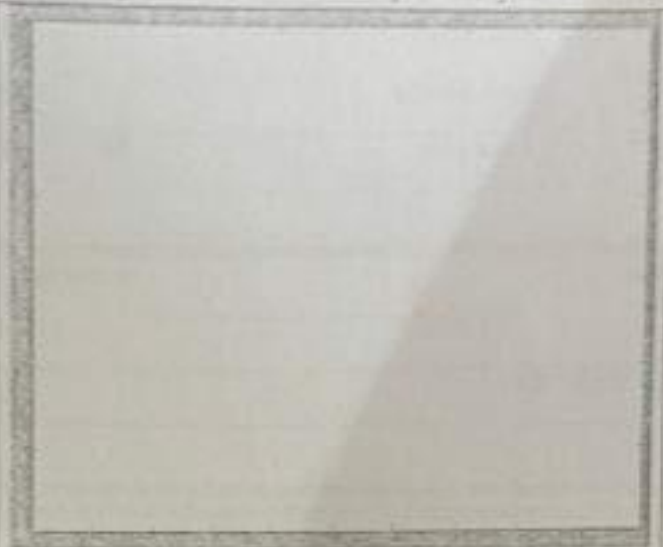
1. $R(t)=1, T=1, A=1$

2. $R(t)=1, T=1, A=0$

3. $R(t)=1, T=3, A=1$

4. $R(t)=1, T=1, A=4$

5. $R(t)=2, T=1, A=1$



Q4) Represent the following system as in Simulink

$$M(t) = 0.5Y''(t) + 2 Y'(t) Z(t) + 3Y(t)$$

$$R(t) = Z''(t) + 4 Z'(t) Y'(t) + 0.6 Z(t)$$